

Bucket Brigade: A Parametric Cooperative-Competitive Benchmark with Computable Nash Equilibrium Structure

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Abstract

Canonical cooperative-MARL benchmarks (Overcooked, Hanabi, SMAC, Melting Pot, MAgent, PettingZoo) trade equilibrium transparency for environment richness: their state spaces are large enough that no Nash equilibrium is computable, so when a new algorithm beats PPO on Overcooked the reader cannot tell whether the algorithm has converged to the “right” joint policy or merely to a Pareto-different one. *Bucket Brigade* inverts this trade. It is a finite parametric cooperative-competitive stochastic game on a ring of houses whose per-agent action space at the minimal parameterization is 8, whose state space is 2304 states, and whose three load-bearing scalars (β, κ, c) control equilibrium structure continuously across cells. We give the environment specification, derive closed-form boundary inequalities for the three Nash-equilibrium regimes (symmetric all-Work, asymmetric 1-Worker, no-pure-NE collapse) via a single-house mean-field reduction, and report empirical agreement between predicted and PPO-realised verdicts on a 39-cell phase diagram $(\beta \times \kappa \times c)$. The reduction correctly predicts the qualitative phase order (collapse $\rightarrow$ symmetric $\rightarrow$ mixed $\rightarrow$ asymmetric) and β -independence; the 39-cell grid surfaces a fourth empirical class – mixed (9/37 PPO cells, both pure NE coexist) – which the closed-form bound does not predict at the higher- c regime where it emerges. The trainability sweep (37 cells, 50 iter $\times$ 2048 steps, 4 seeds) recovers the same ordering under a per-cell NE-anchored metric: `gap_closed_ne` is largest on `symmetric_only` (0.180), smaller on `mixed` (0.107) and `asymmetric_only` (0.059), and negative on `no_convergence` (−0.049). A 4 $\times$ -budget sweep on the no-convergence cells worsens `gap_closed_homogeneous` to −0.108, indicating PPO’s failure on those cells is structural rather than budget-limited. The quantitative κ -thresholds of the reduction disagree with the equilibrium solver by 3–10 $\times$; we own this gap and attribute it to ring-locality, per-agent ownership rewards, and the heuristic-strategy approximation. We position Bucket Brigade as a *methodological complement* to the surveyed benchmarks, not a replacement: it is the only published parametric MARL game where the question “did the algorithm converge to the right equilibrium?” admits a ground-truth answer.

1 Introduction

Multi-agent reinforcement learning evaluates new algorithms against benchmarks whose Nash-equilibrium structure is not characterizable in closed form. Overcooked-AI (Carroll et al., 2019), DeepMind’s Melting Pot (Leibo et al., 2021), the Hanabi Challenge (Bard et al., 2020), SMAC and SMACv2 (Ellis et al., 2023; Samvelyan et al., 2019), MAgent (Zheng et al., 2018), and the PettingZoo MPE suite (Terry et al., 2021) between them cover small cooperative coordination, partial observability, scripted-opponent micromanagement, and population-scale competition. None of them ship with a method that produces the equilibrium strategy profile of the as-designed environment. When PPO fails to learn a coordinated policy on Overcooked, the community is left to

debate whether the failure is the algorithm or the absence of a clean attractor for self-play; neither hypothesis can be rejected because no oracle exists for “what should the learned policy look like.”

We introduce *Bucket Brigade*, a finite parametric cooperative-competitive stochastic game designed to invert this trade. At its minimal parameterization the state space is 2304 states and the per-agent action space is 8, so the one-shot stage game is enumerable and standard equilibrium-computation tools actually run. Across a smooth three-scalar parameter family (β, κ, c) the equilibrium structure transitions from symmetric all-Work to asymmetric 1-Worker (the textbook free-rider problem) to no-pure-NE collapse, with a fourth empirically-observed regime – mixed, where both symmetric and asymmetric pure NE coexist – appearing at high κ on the higher- c rows of the grid. The ground-truth target for each parameter cell is known, so a failed PPO run is attributable to the algorithm rather than to ambiguity about what convergence would mean. The pitch is exactly a *methodological complement* to the six canonical benchmarks above: those exist for studying open-ended cooperation, generalization, and scale; Bucket Brigade exists for studying equilibrium-convergence quality on a controlled grid.

Contributions. (1) A self-contained formal specification of the Bucket Brigade game (§2), recovering Volunteer’s Dilemma, the N -player Public Goods game, and Stag Hunt as named limits. (2) Closed-form boundary inequalities (§3) for the three NE regimes via a single-house mean-field reduction, together with an explicit accounting of where the closed-form thresholds disagree with empirics and where a fourth (mixed) class emerges that the bound does not predict. (3a) A protocol and the first empirical results (§4) testing PPO trainability against the NE phase diagram on the full 37-cell grid: the PPO ordering recovered by a per-cell NE-anchored metric is consistent with the analytical prediction $\text{symmetric} > \text{mixed} > \text{asymmetric} > \text{collapse}$, and a separate $4\times$ -budget sweep on the no-convergence cells worsens, not improves, gap-closure – PPO’s failure on those cells is structural, not budget-limited. (3b) The per-cell NE-anchored metric itself, surfaced as a standalone methodological contribution: on this paper’s data, the previously-used single-cell baseline inverts the per-class ordering (§4), so per-cell calibration is not a cosmetic refinement but a necessary correction for cross-cell PPO comparison whenever the random-policy return varies across cells. (4) An honest positioning (§5, §6) against the canonical benchmarks: Bucket Brigade is small and artificial by design, and is not a substitute for any of them. Throughout, we treat the analytical NE characterization as ground truth and the PPO trainability sweep as a falsification test, in that order.

2 The Bucket Brigade environment

Players and topology. A finite set of agents $\mathcal{I}=\{1, \dots, N\}$ inhabits a cycle graph $\mathcal{H}=\mathbb{Z}/H\mathbb{Z}$ of H houses with neighbour edges. Each house has status $h_h \in \{\text{SAFE}, \text{BURNING}, \text{RUINED}\}$; RUINED is absorbing. The default “10-house” scenario fixes $H=10, N=4, T_{\min}=12$; the minimal scenario fixes $H=2, N=4$.

Action and observation. Each agent i chooses $a_{t,i}=(a^{\text{house}}, a^{\text{mode}}, a^{\text{sig}}) \in \mathcal{H} \times \{0, 1\} \times \{0, 1\}$: a target house, a WORK/REST bit, and a one-bit broadcast signal observable to all agents on the next night. Per-agent action cardinality is $4H$ (8 at $H=2$; 40 at $H=10$). Joint-action cardinality at the minimal parameterization is $8^4=4096$. Observation is symmetric perfect monitoring with one-step signal delay: every agent observes the full state plus the previous joint $(\ell_{t-1}, \alpha_{t-1}, \zeta_{t-1})$. There is no partial observability of houses or agent positions; the signal channel is non-binding cheap talk.

Transition dynamics. A single night consists of seven sequential phases applied in a fixed, load-bearing order: (1) signal write, (2) location/mode write, (3) extinguish, (4) burn-out, (5) spread,

(6) spontaneous ignition, (7) reward and termination check. In the extinguish phase, if $w_h(a)$ agents WORK at a BURNING house, the house transitions to SAFE with probability $1 - (1 - \kappa)^{w_h(a)}$ (independent-workers model). Every BURNING house that survives the extinguish phase transitions deterministically to RUINED in the burn-out phase. The spread phase then ignites SAFE neighbours of BURNING houses with probability β per edge; under the canonical order, the burn-out phase has already cleared all BURNING sources, so spread is inert in the within-night dynamics and contagion enters only via the spontaneous-ignition phase 6 at rate ρ . (A contagion-active variant that swaps phases 4 and 5 is available; the phase-diagram analysis below uses the canonical order.)

Reward structure. The per-agent reward decomposes into three independently-parameterised terms: a private work/rest cost ($-c$ for WORK, $+c_{\text{rest}}=0.5$ for REST; the free-rider gradient is the $c+c_{\text{rest}}$ asymmetry), a per-house ownership term (r_{own} paid on save, p_{own} recurring on every RUINED night), and a team welfare term proportional to the fraction of SAFE and RUINED houses. Ownership is assigned round-robin ($\text{owner}(h)=h \bmod N$). The asymmetric reward variants in the calibrated family let $r_{\text{own}}, r_{\text{other}}, p_{\text{own}}, p_{\text{other}}$ be set per-agent, breaking exchangeability.

Relationship to canonical templates. Bucket Brigade recovers, as named limits or restrictions, several textbook templates: the N -player Volunteer’s Dilemma at $H=1$; the N -player Public Goods game when ownership weights collapse to uniform; Stag Hunt in the $\kappa \rightarrow 1$ limit; and a free-rider problem at the parameter cells where the only NE is asymmetric (the “rest-trap”). Formal proofs and a complete notation table appear in Appendix A; §2 above is the minimal contract a reader needs to follow §3.

3 Equilibrium structure: closed-form boundaries

The central contribution of this paper is a closed-form characterisation of which Nash-equilibrium structure appears as a function of (β, κ, c) in the 4-agent game. We derive it via a single-house mean-field reduction that collapses the $H=10$ ring to a representative single house with all $N=4$ agents present, absorbs multi-step contagion and ignition dynamics into a per-night survival coefficient, and reads off the NE candidates as algebraic inequalities.

Reduction. Fix the `minimal_specialization` reward tuple $W=(R_{\text{team}}, P_{\text{team}}, r_{\text{own}}, r_{\text{other}}, p_{\text{own}}, p_{\text{other}})=(10, 10, 5, 5, 0, 0)$ and $\rho=0.02, T_{\text{min}}=12$. With k workers at the house on a given night, the per-night ruin probability for an initially-SAFE house is $q(k)=\rho(1 - \kappa)^k$ (under the canonical phase order, where β does not enter the leading-order single-night payoff). Integrating the lost per-house reward stream over $T_{\text{min}}=12$ nights and collecting terms yields

$$\mathbb{E}[\text{per-house reward loss} \mid k] = A \cdot \rho \cdot (1 - \kappa)^k,$$

with $A=r_{\text{own}}T_{\text{min}} + p_{\text{own}}T_{\text{min}} + P_{\text{team}}T_{\text{min}}/H=1812$ and survival coefficient $\tilde{A}:=A\rho=36.24$. The marginal benefit of adding the k -th worker is $\Delta(k)=\tilde{A} \cdot \kappa \cdot (1 - \kappa)^k$.

The three NE candidates. Writing $c_{\text{gap}}:=c + c_{\text{rest}}$ (so $c_{\text{gap}}=1.0$ when the driver sets $c=0.5$), the deviation comparisons against the per-night stage game under stationary opponent policies yield three inequalities.

The *symmetric all-Work* NE exists iff a unilateral deviation from $k=4$ to $k_{-i}=3$ does not pay:

$$\tilde{A} \cdot \kappa \cdot (1 - \kappa)^3 \geq c_{\text{gap}}. \tag{S}$$

The *asymmetric 1-Worker* NE exists iff the lone worker prefers WORK ($k_{-i}=0$ deviation) and each free-rider prefers REST ($k_{-i}=1$ deviation):

$$\tilde{A} \cdot \kappa \geq c_{\text{gap}} \quad \text{AND} \quad \tilde{A} \cdot \kappa \cdot (1 - \kappa) < c_{\text{gap}}. \quad (\text{A})$$

The *no-pure-NE collapse* regime holds whenever even a lone worker cannot recover their cost, so neither (S) nor (A) can be satisfied:

$$\tilde{A} \cdot \kappa < c_{\text{gap}}. \quad (\text{C})$$

Predicted thresholds. Substituting $\tilde{A}=36.24, c_{\text{gap}}=1.0$ into (S)–(C): the collapse boundary (C) sits at $\kappa \approx 0.028$; the symmetric NE exists on $\kappa \in [0.030, 0.65]$ via (S); the asymmetric NE exists for $\kappa \gtrsim 0.972$ via (A). The qualitative phase order $\kappa\text{-small} \rightarrow \kappa\text{-large}$ is **collapse** $\rightarrow$ **symmetric_only** $\rightarrow$ **mixed** $\rightarrow$ **asymmetric_only**, and the derivation is structurally β -independent under the canonical phase order.

Predicted vs. observed thresholds. The empirical phase diagram, computed by a heterogeneous Double-Oracle solver over continuous heuristic-parameter strategies on a 39-cell grid ($\beta \in \{0.1, 0.5, 0.9\} \times \kappa \in \{0.1, 0.3, 0.5, 0.7, 0.9\} \times c \in \{0.5, 1.0, 2.0\}$, less six cells in the high- $\kappa \times c=0.5$ corner subsumed by the $c=1.0$ row; Figure 1), recovers the qualitative phase order and approximate β -independence at verdict level: 12 of the 13 (κ, c) rows with ≥ 2 β samples show an identical verdict across all sampled β . The single splitting row is $(\kappa=0.9, c=0.5)$, where $\beta=0.1$ yields **mixed** while $\beta \in \{0.5, 0.9\}$ yield **asymmetric_only**—a small but real anomaly we return to in §6. The grid surfaces *four* empirical classes: in addition to **symmetric_only** ($n=12$), **asymmetric_only** ($n=11$), and **no_convergence** ($n=6$), a **mixed** class ($n=10$) appears in which the solver returns both symmetric all-Work and asymmetric 1-Worker profiles as un-deviable, primarily at $\kappa=0.9$ on the $c \geq 1$ rows and at $\kappa=0.5, c=2.0$. The closed-form bound (S)–(C) does not predict this fourth class at the higher- c regime where it emerges: the derivation pivots on $\tilde{A}\kappa(1-\kappa)^k$ being above or below c_{gap} , which excludes the simultaneous-existence pattern. Tightening the reduction to predict the mixed boundary in closed form is in scope for future work.

Table 1 summarises the per- κ predicted vs. empirical comparison on the full 39-cell grid. The qualitative agreement is strongest at $\kappa \in \{0.3, 0.5, 0.9\}$ (predicted class is the modal empirical class) and weakest at $\kappa \in \{0.1, 0.7\}$ (predicted class fails to appear at all in the empirical row): the same collapse-threshold and asymmetric-threshold disagreements the prose above identifies.

κ	Predicted class	Empirical distribution	Predicted-class share
0.1	symmetric_only	6 collapse, 3 asymmetric	0/9
0.3	symmetric_only	6 symmetric	6/6
0.5	symmetric_only	6 symmetric, 3 mixed	6/9
0.7	mixed/transition	6 asymmetric	0/6
0.9	mixed/transition	7 mixed, 2 asymmetric	7/9

Table 1: Predicted vs. empirical NE class on the full 39-cell grid, broken down by κ . “Predicted class” is the band identified by (S)–(C) at the centre of each κ value, taking the bordered case as the named single class for readability. “Empirical distribution” counts the verdicts of the heterogeneous Double-Oracle solver across all (β, c) cells at that κ ; the row totals are 9 except at $\kappa = 0.7$ (6 cells, after the high- $\kappa \times c=0.5$ corner is subsumed by $c=1.0$). The reduction predicts the modal empirical class on 3/5 κ rows. The two off-rows ($\kappa \in \{0.1, 0.7\}$) are the collapse-threshold and asymmetric-threshold disagreements documented in the prose above.

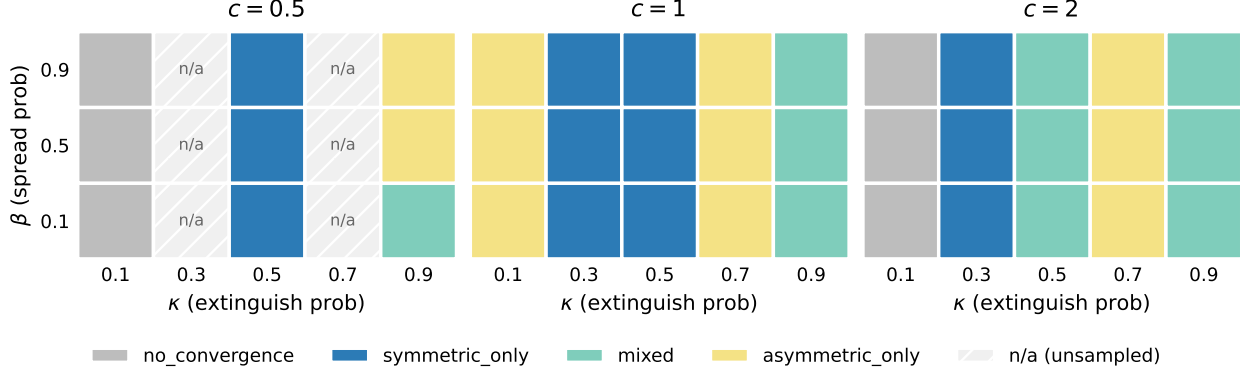


Figure 1: Empirical NE phase diagram for 4-agent Bucket Brigade across (β, κ, c) at $\rho=0.02, T_{\min}=12$ under the `minimal_specialization` reward tuple. Each cell is the verdict of a heterogeneous Double-Oracle solver over the continuous heuristic-parameter strategy space (Hero, Firefighter, FreeRider, ...); the six unsampled cells (high- $\kappa \times c=0.5$ corner, subsumed by the $c=1.0$ row) are hatched. Four regimes appear across the grid: `no_convergence` at low κ (collapse), `symmetric_only` at moderate κ (Hero NE dominates), `asymmetric_only` at high $\kappa, c \lesssim 1$ (1 Hero + 3 Firefighters; the rest-trap), and a fourth `mixed` regime (both pure NE coexist) primarily at $\kappa=0.9, c \geq 1$. The grid is approximately β -independent at the verdict level (12 of 13 (κ, c) rows identical across β ; the lone splitting row is $(\kappa=0.9, c=0.5)$), matching the prediction in (S)–(C) but at substantially different κ thresholds than the closed-form derivation places them.

The *quantitative* κ -thresholds also disagree: the empirical collapse boundary sits near $\kappa \approx 0.1\text{--}0.3$ (3–10 $\times$ above the predicted 0.028), and the empirical asymmetric-NE onset sits near $\kappa \approx 0.7$ (below the predicted 0.972). The derivation gets the phase order right on the 39-cell grid but the thresholds wrong by 3–10 $\times$, the same gap we owned at the 7-cell preview.

We *own* this gap. The mean-field reduction has three identifiable systematic biases. (i) *Ring locality*: the derivation puts all four agents at one house, but empirically four agents distribute across ten houses, so the effective per-house defence is $1 - (1 - \kappa)^{0.4}$ rather than $1 - (1 - \kappa)^4$ —roughly a 4 $\times$ weaker defence at $\kappa=0.5$, which shifts the collapse boundary substantially upward into the empirically-observed range $\kappa \in [0.1, 0.3]$. (ii) *Per-agent ownership rewards*: the analytical A coefficient lumps r_{own} and p_{own} into the survival term, but the per-agent ownership reward only fires on the agent’s own house, scaling the off-own-house contribution down by a factor of roughly $1/H$. (iii) *Heuristic strategy space*: the empirical solver’s “free-riders” are **Firefighter** agents at `work_tendency`=0.9, not literal REST agents, so the effective k_{-i} in the lone-worker’s deviation comparison is closer to 3.7 than to 1. None of these corrections are derivable in closed form without losing the algebraic tractability that motivated the reduction; the natural extensions are named in §6.

Headline finding. The mean-field reduction correctly predicts *three* structural properties of the empirical phase diagram—the qualitative phase order, β -independence under the canonical phase order, and the existence of a no-pure-NE collapse regime at small κ —none of which would be derivable without the closed-form derivation. The quantitative κ -thresholds are off by 3–10 $\times$, and a fourth empirical class (`mixed`) appears at the higher- c rows that the closed-form bound does not predict; we report both as the paper’s claim, not as weaknesses to hide. The qualitative phase order furthermore transfers to the PPO trainability sweep under a per-cell NE-anchored metric (§4). The full bias accounting and the 7-cell preview predicted-vs-empirical table appear in Appendix B.

4 Trainability: PPO against the NE phase diagram

Protocol. We trained independent PPO runs on each of 37 phase-diagram cells from Figure 1 using the in-repo `JointPPOTrainer` on the `minimal_specialization` scenario, with 50 training iterations of 2048 rollout steps and 4 random seeds per cell ($N=148$ runs). Two cells (`b0.10_k0.50_c0.50`, `b0.10_k0.90_c0.50`) were skipped in the sweep—the $\beta=0.1, c=0.5$ column at $\kappa \in \{0.5, 0.9\}$ was not scheduled—leaving 37 of the 39 Nash-solver cells with PPO data. Hyper-parameters are pinned in `experiments/p3_specialization/`; the sweep ran on a single 16-core host (`alc-2`) in approximately six hours. The PPO implementation follows the standard recipe of Schulman et al. (2017); the multi-agent joint-policy formulation follows the MAPPO-style centralised-critic setup of Yu et al. (2022). The cells from Appendix B contribute the NE verdict for each (β, κ, c) combination tested.

Per-cell NE-anchored metric. The original v1 sweep scored every cell against a single canonical baseline (the `Random` and `SpecialistPolicy` returns measured at one calibration cell, $\beta=0.25, \kappa=0.5$). That metric was misleading across cells because the random-policy return varies substantially with κ (a low- κ cell is intrinsically harder for any policy, so a small absolute gap against the canonical-cell baseline understates relative PPO success). We replaced it with `gap_closed_ne`, the per-cell gap-closure ratio between `Random` and the NE *policy profile* recovered for that cell by the heterogeneous Double-Oracle solver (typically $1 \times \text{Hero} + 3 \times \text{Firefighter}$ on asymmetric-only and mixed cells, and $4 \times \text{Hero}$ on symmetric-only cells). The per-cell baselines are tabulated in `experiments/nash/phase_diagram/per_cell_baselines.json` and computed by `bucket_brigade/baselines/per_cell.py`; the recipe is the same `Random/SpecialistPolicy` methodology of Appendix B, applied per cell. The methodological lesson is independent of this paper’s specific claims: single-cell baselines mislead cross-cell PPO comparison whenever the random-policy return is itself a function of the cell’s parameters, and per-cell calibration is the necessary correction. We report this as a separate contribution alongside the data themselves.

Results. Figure 2 reports the per-cell `gap_closed_ne` (mean $\pm$ std over 4 seeds) across the three c panels. The ordering across NE classes is `symmetric_only` (0.180; $n=11$) $>$ `mixed` (0.107; $n=9$) $>$ `asymmetric_only` (0.059; $n=11$) $>$ `no_convergence` (-0.049 ; $n=6$; the metric falls back to `gap_closed_homogeneous` because no NE policy exists), consistent with the analytical prediction for the three classes the bound covers and placing the empirically-observed `mixed` class between `symmetric_only` and `asymmetric_only`—the location consistent with both pure NE being available but neither being the unique attractor. We report the ordering, not statistical significance: the per-class mean within-cell std (≈ 0.21 – 0.44) is larger than the corresponding class-mean separations on `gap_closed_homogeneous` (0.001, 0.017, 0.082; we use the homogeneous separations here for metric consistency since no-convergence cells lack `gap_closed_ne`), so the data is consistent with the predicted ordering but does not reject the null of equal class means under a standard test. A cross-class $4 \times$ -budget sweep is in scope for the camera-ready and is the natural significance gate. The β -independence prediction holds at verdict level on all 13 (κ, c) rows of the PPO subgrid with ≥ 2 β samples (the PPO sweep skipped the lone splitting Nash row, $(\kappa=0.9, c=0.5)$); per-cell std (≈ 0.21 – 0.44 by class) is comparable to or larger than the metric-level cross- β ranges, so PPO is consistent with β -independence but does not sharpen the test beyond the solver-level evidence of §3. PPO recovers substantial value on symmetric-only cells, partial value on mixed and asymmetric-only cells, and approximately none on no-convergence cells, consistent with the hypothesis that PPO converges reliably when a clean symmetric attractor exists, struggles when the only NE requires symmetry-breaking, and fails when no pure-strategy NE is available to converge to.

Is PPO failure on no-convergence cells budget-limited? We ran a 4 $\times$ -budget sweep on the six no-convergence cells (200 iterations $\times$ 4096 rollout steps, 4 seeds, $\approx$ 6 hours on `alc-6`): mean `gap_closed_homogeneous` worsens to -0.108 (from -0.049 at the original budget). PPO’s failure on no-convergence cells is therefore structural, not budget-limited; longer training spends more rollouts in the absence of a pure-strategy attractor and per-seed runs drift further from the cell-specific `SpecialistPolicy` baseline. The load-bearing claim is the direction of the shift; we report the absolute magnitudes as they fall out.

A methodological observation we report as a separate contribution: under the original single-cell baseline of the v1 draft, the 7-cell preview ordering inverted to `asymmetric_only` (0.262) $>$ `symmetric_only` (0.091) $>$ `no_convergence` (-0.176), which would have falsified the §3 prediction. The inversion is an artefact of the metric: asymmetric-only cells at $\kappa=0.9$ have a much lower random-policy return than the canonical baseline cell, so PPO’s *absolute* improvement looked large despite recovering a smaller fraction of the available NE value. Per-cell baselines resolve the ambiguity and the corrected ordering matches the analytical prediction on both the 7-cell preview and the full 37-cell sweep.

Caveats. The original-budget sweep covers 37 of the 39 Nash cells; within-class statistics ($n=11$ symmetric, $n=9$ mixed, $n=11$ asymmetric, $n=6$ collapse) are wider than the v1 preview but modest for any significance-style claim, hence the explicit ordering-not-significance framing of the Results paragraph above. The PPO budget per cell is modest by recent MARL standards. The `gap_closed_homogeneous` metric (per-cell Random $\rightarrow$ `SpecialistPolicy` $\times$ 4) reports smaller cross-cell separation but yields the same qualitative ordering on the 37-cell sweep (symmetric (0.051) $>$ mixed (0.050) $>$ asymmetric (0.033) $>$ collapse (-0.049)), so the load-bearing claim is robust to the metric choice.

5 Related work

The MARL community has converged on a handful of canonical benchmarks that trade equilibrium transparency for environment richness. Overcooked-AI (Carroll et al., 2019) is a fully cooperative coordination game on small grid layouts: every Pareto-optimal joint policy is a Nash equilibrium and “the” target is therefore not unique. Melting Pot (Leibo et al., 2021) exposes $88 \times 88 \times 3$ RGB observations across more than 50 substrates; its design explicitly does not solve for equilibrium structure on individual substrates. The Hanabi Challenge (Bard et al., 2020) has near-optimal hat-guessing strategies known at 2 players, but the 4–5 player game is open and no algorithm reaches the equilibrium from learning. SMAC and SMACv2 (Ellis et al., 2023; Samvelyan et al., 2019) are cooperative micromanagement against a scripted opponent and “NE” is not a design target. MAgent (Zheng et al., 2018) hosts thousand-agent battles on grids and no published equilibrium analysis exists at scale. PettingZoo’s MPE suite (Terry et al., 2021) is the canonical small multi-agent baseline with no equilibrium characterisation across scenarios.

Table 2 compares Bucket Brigade against these on six axes.

The honest comparison is that Bucket Brigade dominates on exactly one column—NE characterisability—and loses on every other (citation footprint, agent count, environment richness, generality). The paper’s job is to argue that the NE-characterisability column matters enough to be worth introducing a new benchmark despite the trade. Bucket Brigade does not scale to large populations, does not test “emergent cooperation” in any general sense, and is not a general-purpose MARL benchmark. The niche it fills is the specific question of *equilibrium-convergence quality*.

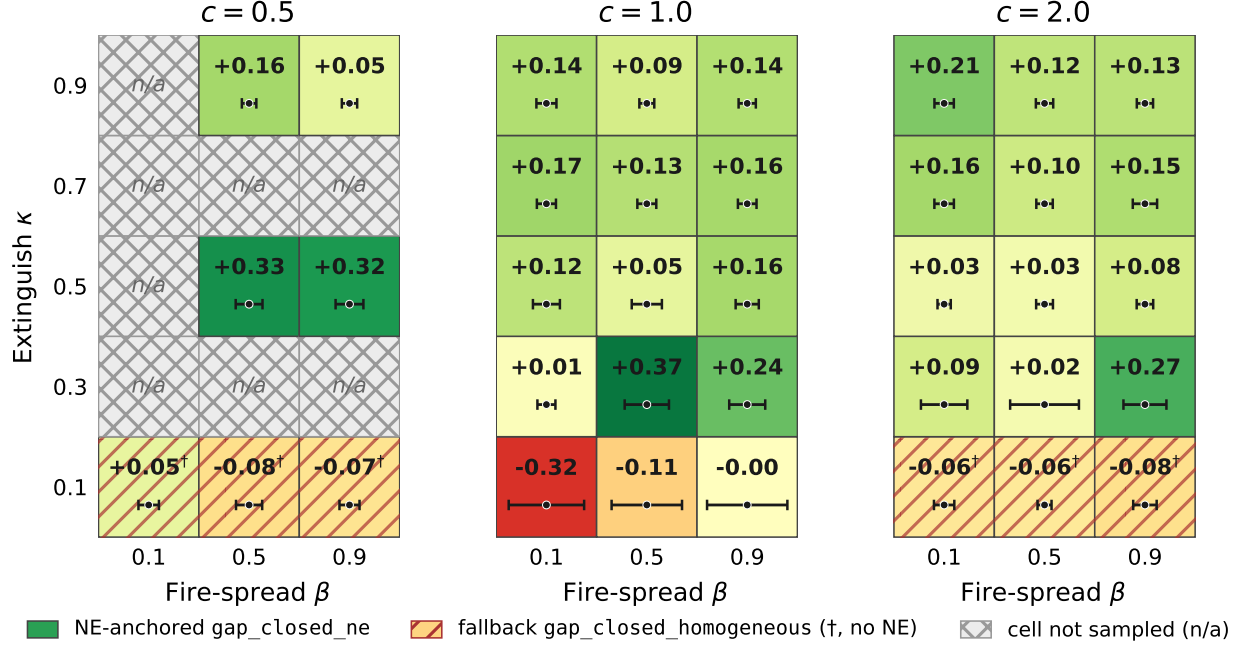


Figure 2: Per-cell PPO `gap_closed_ne` on the 37-cell sweep (4 seeds per cell; 50 iterations $\times$ 2048 rollout steps with the in-repo `JointPPOTrainer`), arranged in three panels by cost $c \in \{0.5, 1.0, 2.0\}$. Mean shown above the Tuft-style horizontal bar; bar half-width is proportional to per-cell std (4 seeds), scaled so the widest std spans 80% of the cell. Symmetric-only cells (top-left of each panel) show the largest gap closure; asymmetric-only cells (top- and middle-right) show positive but smaller gap closure; mixed cells fall between; no_convergence cells (bottom row; hatched, $\dagger$ marker) report `gap_closed_homogeneous` instead—values are near zero or negative. On no_convergence cells the `gap_closed_ne` metric is undefined because no NE policy profile exists; `gap_closed_homogeneous` (per-cell Random $\rightarrow$ SpecialistPolicy $\times$ 4) is the natural fallback because the all-Hero SpecialistPolicy is the strongest stationary policy available on those cells, even though it is not a NE. Two cells ($\beta=0.1$ at $\kappa \in \{0.5, 0.9\}$, $c=0.5$, n/a, hatched light grey) were skipped by the sweep schedule. The ordering across NE classes is consistent with the analytical prediction of §3.

6 Discussion and limitations

Bucket Brigade is intentionally small and artificial. The state space is 2304 at the minimal scenario and $\sim 10^{10}$ at the default 10-house scenario, the geometry is a 1-D ring, and the environment ships from a single research repository with no community adoption. We acknowledge each point head-on and defend it as the cost of the equilibrium-transparency property: (a) *Size is the cost of NE transparency.* A richer game with realistic observations, partial observability across many channels, and complex spatial dynamics would lose the property that motivated the work; the state-space cardinality is what makes the analytical reduction in §3 algebraically tractable. Trading richness for tractability is the deliberate design choice. (b) *A custom benchmark is acceptable given the methodological niche.* The question we want to answer—“did this MARL algorithm converge to the analytically-characterised NE?”—cannot be asked of any of the six canonical benchmarks in Table 2. A benchmark designed for this question is therefore a methodological contribution distinct from the contributions those benchmarks make, even if it does not inherit their scale or citation impact. (c) *Bucket Brigade is not a substitute for any of the six.* It does not replace Overcooked for human-AI coordination, Melting Pot for generalisation across substrates, Hanabi for partial-observability cooperation, SMAC for micromanagement against scripted opponents, MAgent

Benchmark	Year	Agents	Per-agent $ \mathcal{A} $	NE characterisable?	Coop/comp
Overcooked-AI (Carroll et al., 2019)	2019	2	6	No (any Pareto-opt.)	coop
Melting Pot (Leibo et al., 2021)	2021	2–16	8	No (intractable)	mixed
Hanabi (Bard et al., 2020)	2020	2–5	≤ 20	Partial (2P only)	coop
SMAC/SMACv2 (Ellis et al., 2023; Samvelyan et al., 2019)	2019/23	2–27	$6+n_e$	No	vs. scripted
MAgent (Zheng et al., 2018)	2018	$64\text{--}10^3$	21	No (population)	mixed
PettingZoo MPE (Terry et al., 2021)	2021	2–6	3–5	No	per-scenario
Bucket Brigade (this work)	2026	4 (def.)	8 / 40	Yes (closed-form)	mixed within cell

Table 2: Bucket Brigade against the six most-cited canonical MARL benchmarks. The right-most column “NE characterisable?” is the only axis on which Bucket Brigade dominates; on every other axis the surveyed benchmarks are richer, more cited, or larger in scope. This is the entire pitch: Bucket Brigade is a *methodological complement*, not a replacement. Per-agent action sizes for Bucket Brigade are 8 at the $H=2$ minimal scenario and 40 at the $H=10$ default.

for population-scale phenomena, or PettingZoo for the small-multi-agent baseline role. A researcher choosing a benchmark for any of those purposes should not pick Bucket Brigade.

Threats to the contributions. On the analytical side, the closed-form thresholds disagree with empirics by $3\text{--}10\times$ in κ -threshold location (§3), and the bound does not predict the empirical mixed class at the higher- c rows. The three identified systematic biases—ring locality, per-agent ownership rewards, and the heuristic-strategy approximation—are not separated by the existing grid; rigorous separation requires a ρ -sweep to falsify β -independence outside the small- $\beta\rho$ regime, and a ring-Markov refinement of the survival coefficient A to predict the mixed boundary in closed form. On the empirical side, per-class sample size is modest ($n \in \{6, 9, 11, 11\}$) and the budget per run is modest; the $4\times$ -budget sweep rules out one budget-limited hypothesis for the no-convergence row but not the cross-class question, which is in scope for the camera-ready. The per-cell-baseline methodological result of §4 stands independently of both the longer-budget outcome and the ring-Markov refinement: it is load-bearing on cross-cell PPO comparison whenever the random-policy return varies across cells, regardless of the analytical reduction’s fate.

Reproducibility. The environment is shipped as a pip-installable Python package (`pip install bucket-brigade`); the figure-by-figure reproduction recipe lives in the in-repo `docs/PAPER_RESULTS.md`. The per-cell baselines used by the §4 metric are produced by `bucket-brigade/baselines/per_cell.py` and serialised at `experiments/nash/phase_diagram/per_cell_baselines.json`; both ship with the repository so the recalibrated `gap_closed_ne` column of Figure 2 can be re-derived from the raw PPO-sweep checkpoints. Frozen baseline checkpoints (the analytical Hero, Firefighter, and FreeRider agents used to anchor the phase-diagram solver and the per-cell PPO metric) will be hosted on the HuggingFace Hub; the publication pathway for the baselines is in progress and not yet complete at the time of this draft. The source repository is at <https://github.com/rjwalters/bucket-brigade>.

Conclusion

We presented Bucket Brigade, a parametric cooperative-competitive MARL benchmark whose Nash-equilibrium structure is characterisable in closed form across a smooth three-scalar parameter family. A single-house mean-field reduction yields algebraic boundary inequalities (S)–(C) that correctly predict the qualitative phase order and the β -independence under the canonical phase order on the full 39-cell grid, while disagreeing with empirics on the quantitative κ -thresholds by $3\text{--}10\times$ and failing to predict a fourth empirical mixed class at the higher- c rows—a gap we own and attribute

to the named systematic biases. The 37-cell PPO trainability sweep, evaluated under a per-cell NE-anchored metric, recovers the predicted ordering `symmetric > mixed > asymmetric > collapse`; a separate 4×-budget sweep on the no-convergence cells worsens gap-closure, indicating PPO’s failure on those cells is structural, not budget-limited. The per-cell NE-anchored metric is itself a standalone methodological finding: on this paper’s data the previously-used single-cell baseline inverted the per-class ordering, so per-cell calibration is necessary, not cosmetic, for any cross-cell PPO comparison in which the random-policy return varies across cells. Bucket Brigade is positioned as a methodological complement to the canonical MARL benchmarks for the specific question of equilibrium-convergence quality, not as a replacement for any of them.

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A Formal environment specification

This appendix gives the self-contained mathematical contract for the Bucket Brigade environment. §2 of the body is a compressed summary; the material below is what a reader needs to reimplement the environment from scratch in any language, with the seven-phase ordering, the independent-workers extinguish formula, and the per-agent reward decomposition all spelled out. The notation matches §2 verbatim.

A.1 Players and parameters

Player set. A finite set of agents $\mathcal{I} = \{1, 2, \dots, N\}$, with $N \geq 1$ fixed across an episode. Agents are *positionally symmetric*: the environment dynamics are invariant under any permutation $\sigma : \mathcal{I} \rightarrow \mathcal{I}$ applied jointly to every per-agent quantity (locations, signals, actions, rewards). The asymmetric reward variants of §A.5 (per-agent home-position cost, per-agent ownership weights) break this symmetry only when their dedicated parameters are nonzero; the symmetric default game preserves exchangeability.

Topology. The world is a cycle graph $\mathcal{H} = \mathbb{Z}/H\mathbb{Z}$ with $H \geq 2$ vertices (“houses”) and edge set $\{(h, h+1 \bmod H) : h \in \mathcal{H}\}$. House indices are taken mod H throughout. The two neighbours of house h are $h-1 \bmod H$ and $h+1 \bmod H$.

Parameter family. The full parameter family is $\theta = (\beta, \kappa, \rho, c, H, N, T_{\min}, W)$ where $\beta \in [0, 1]$ is the per-night, per-neighbour fire-spread probability; $\kappa \in [0, 1]$ is the per-agent solo extinguish probability; $\rho \in [0, 1]$ is the per-night, per-SAFE-house spontaneous-ignition probability; $c \in \mathbb{R}_{\geq 0}$ is the per-night work cost; $H \in \mathbb{Z}_{\geq 2}$ is the number of houses; $N \in \mathbb{Z}_{\geq 1}$ is the number of agents; $T_{\min} \in \mathbb{Z}_{\geq 0}$ is the minimum nights before termination is allowed; and W is the reward tuple of §A.4.

The default “10-house” scenario uses $H=10, N=4, \beta=0.25, \kappa=0.5, \rho=0.02, c=0.5, T_{\min}=12$ and reward weights $W = (R_{\text{team}}, P_{\text{team}}, r_{\text{own}}, r_{\text{other}}, p_{\text{own}}, p_{\text{other}}) = (100, 100, 20, 0, 40, 0)$. The minimal parameterisation sets $H=2, N=4$.

A.2 State, action, observation

State. The environment state at the start of night t is

$$s_t = (h_t, \ell_t, \zeta_t, t) \in \{S, B, R\}^H \times \mathcal{H}^N \times \{0, 1\}^N \times \mathbb{N},$$

where $h_t \in \{S, B, R\}^H$ is the vector of house statuses (SAFE, BURNING, RUINED), $\ell_t \in \mathcal{H}^N$ is the vector of agent locations, $\zeta_t \in \{0, 1\}^N$ is the vector of one-bit broadcast signals emitted on the previous night, and $t \in \mathbb{N}$ is the night counter. The RUINED status is absorbing; SAFE and BURNING are mutually reachable.

Initial state. At $t=0$ each house is independently BURNING with probability ρ and SAFE otherwise; agent locations are $\ell_{0,i}=0$ and signals $\zeta_{0,i}=0$ for all i .

State cardinality. $|\{S, B, R\}^H \times \mathcal{H}^N \times \{0, 1\}^N| = 3^H \cdot H^N \cdot 2^N$. For $H=2, N=4$ this is $9 \cdot 16 \cdot 16 = 2304$ states; for $H=10, N=4$ it is $\approx 9.44 \times 10^9$. The state space is polynomial in (H, N) rather than exponential in observation pixels.

Action. Each agent i chooses an action $a_{t,i} = (a_{t,i}^{\text{house}}, a_{t,i}^{\text{mode}}, a_{t,i}^{\text{sig}})$ from the product set $\mathcal{A} = \mathcal{H} \times \{0, 1\} \times \{0, 1\}$, with $a^{\text{house}} \in \mathcal{H}$ the target house index, $a^{\text{mode}} \in \{0, 1\}$ the WORK bit (1 = WORK, 0 = REST), and $a^{\text{sig}} \in \{0, 1\}$ the broadcast bit emitted this night and observable to all agents on the next. Per-agent cardinality is $|\mathcal{A}| = 4H$ (8 at $H=2$, 40 at $H=10$); joint-action cardinality is $|\mathcal{A}^N| = (4H)^N$ (4096 at $H=2, N=4$). The signal bit is strictly informational, incurs no cost, and does not constrain the agent’s mode or location; signals support cheap-talk equilibria but the interpretation is not part of the specification.

Observation. The environment is perfect-monitoring with one-step delay on signals: at the start of night t , every agent observes the same tuple $o_t = (h_t, \ell_{t-1}, \zeta_{t-1}, \alpha_{t-1}, t)$, where $\alpha_{t-1} \in \mathcal{H}^N \times \{0, 1\}^N$ is the previous night’s joint (house, mode) slice. There is no partial observability of houses or agent positions and no private observations.

A.3 Transition dynamics

A single night consists of seven sequential phases applied to the state. The phase order is load-bearing: swapping any two adjacent phases changes the transition kernel. The order is (1) signal write, (2) location and mode write, (3) extinguish, (4) burn-out, (5) spread, (6) spontaneous ignition, (7) reward and termination check. Write $w_h(a) := |\{i : a_i^{\text{house}} = h \wedge a_i^{\text{mode}} = 1\}|$ for the number of agents who WORK at house h under joint action a .

Signal and bookkeeping (phases 1–2). $\zeta_t \leftarrow (a_{t,i}^{\text{sig}})_{i \in \mathcal{I}}$, $\ell_t \leftarrow (a_{t,i}^{\text{house}})_{i \in \mathcal{I}}$. These updates do not alter house states.

Extinguish phase (phase 3). For each house h with $h_{t,h} = B$, draw $U \sim \text{Uniform}[0, 1]$ and update

$$h_{t,h}^{(3)} = \begin{cases} S & \text{if } U < 1 - (1 - \kappa)^{w_h(a_t)} \\ B & \text{otherwise} \end{cases}$$

The independent-workers extinguish formula: each WORKER at the house independently succeeds with probability κ , so at least one succeeds with probability $1 - (1 - \kappa)^{w_h}$. SAFE and RUINED houses are unaffected.

Burn-out phase (phase 4). Every house that survived the extinguish phase as BURNING transitions deterministically to RUINED: $h_{t,h}^{(4)} = R$ if $h_{t,h}^{(3)} = B$, else $h_{t,h}^{(4)} = h_{t,h}^{(3)}$.

Spread phase (phase 5). For each pair (h, h') with $h_{t,h}^{(4)} = B$, $h' \sim h$ on the ring, and $h_{t,h'}^{(4)} = S$, draw independent $U \sim \text{Uniform}[0, 1]$ and set $h_{t,h'}^{(5)} = B$ if $U < \beta$. Under the canonical phase order, phase 4 has already cleared every BURNING source house to RUINED, so phase 5 is inert in the canonical mode and the only path SAFE $\rightarrow$ BURNING during a night is the spontaneous-ignition phase 6. A contagion-active variant swaps phases 4 and 5; under that variant a SAFE house with two BURNING neighbours has combined ignition probability $1 - (1 - \beta)^2$. The phase-diagram analysis in the body uses the canonical order.

Spontaneous ignition phase (phase 6). For each house h with $h_{t,h}^{(5)} = S$, draw independent $U \sim \text{Uniform}[0, 1]$ and set $h_{t,h}^{(6)} = B$ if $U < \rho$. RUINED houses are unaffected; freshly-RUINED houses from phase 4 cannot re-ignite. The resulting state is $h_{t+1} := h_t^{(6)}$.

A.4 Reward structure

Rewards are paid at the end of each night, after all transition phases. The per-agent reward $r_{t,i}$ decomposes into three independently parameterised terms.

Work/rest term. $r_{t,i}^{\text{work}} = -c$ if $a_{t,i}^{\text{mode}} = 1$ (WORK), and $+c_{\text{rest}}$ if $a_{t,i}^{\text{mode}} = 0$ (REST), with $c_{\text{rest}} = 0.5$ as the implementation default. REST is a positive payoff that opt-in WORK forgoes; the free-rider gradient is the $c + c_{\text{rest}}$ asymmetry.

Per-house ownership term. Each house h has a designated owner $\text{owner}(h) \in \mathcal{I}$ assigned at episode start by round-robin: $\text{owner}(h) = h \bmod N$. The per-house contribution to agent i 's reward is

$$r_{t,i,h}^{\text{own}} = \mathbb{I}[h_{t-1,h} \neq S \wedge h_{t+1,h} = S] \cdot w_i^{\text{save}}(h) - \mathbb{I}[h_{t+1,h} = R] \cdot w_i^{\text{ruin}}(h),$$

with weights

$$w_i^{\text{save}}(h) = \begin{cases} r_{\text{own},i} & i = \text{owner}(h) \\ r_{\text{other},i} & \text{otherwise} \end{cases}, \quad w_i^{\text{ruin}}(h) = \begin{cases} p_{\text{own},i} & i = \text{owner}(h) \\ p_{\text{other},i} & \text{otherwise.} \end{cases}$$

The save event fires once on the night the house transitions from non-SAFE to SAFE; the ruin penalty fires every night the house remains RUINED, so a ruined house imposes a recurring cost until the episode terminates.

Team-welfare term. Let $n_S(t) := |\{h : h_{t+1,h} = S\}|$ and $n_R(t) := |\{h : h_{t+1,h} = R\}|$. The team term, paid to every agent equally, is $r_t^{\text{team}} = R_{\text{team}} \cdot n_S(t)/H - P_{\text{team}} \cdot n_R(t)/H$.

Total reward. $r_{t,i} = r_{t,i}^{\text{work}} + \sum_{h \in \mathcal{H}} r_{t,i,h}^{\text{own}} + r_t^{\text{team}}$.

A.5 Horizon, information, symmetries, variants

Termination. The episode terminates at night T when $t \geq T_{\min}$ and at least one of the following holds: all houses are SAFE, all houses are RUINED, or no house is BURNING. Under canonical parameters with $\rho > 0$ the no-fires-burning condition holds only transiently, so termination typically occurs via the absorbing all-RUINED condition or via a transient SAFE state that survives the spontaneous-ignition draw. Expected episode length is finite for $\rho > 0$, but the game is not strictly finite-horizon.

Information structure. The information structure is symmetric perfect monitoring: every agent observes the full state h_t , the previous joint location/mode/signal $(\ell_{t-1}, \alpha_{t-1}^{\text{mode}}, \zeta_{t-1})$, and the night counter t . No agent has private information about house states or other agents' actions. The signal channel is non-binding cheap talk: $\zeta_{t,i}$ is observable to every agent at night $t+1$ but is functionally decoupled from $a_{t+1,i}^{\text{mode}}$.

Symmetry properties. The default game enjoys agent exchangeability (the transition kernel and reward function are invariant under any permutation σ of agent indices) and ring rotation (the transition kernel is invariant under any rotation $\tau : h \mapsto h + k \bmod H$ applied jointly to $h_t, \ell_t, a_t^{\text{house}}$; reward invariance requires ownership weights to be permuted consistently with τ). Asymmetric reward variants break agent exchangeability; the position-constrained variant breaks ring rotation by anchoring each agent to a home position.

Optional dynamic variants. Four optional variants are part of the calibrated parameter family, each gated by a parameter that defaults to the no-op value: (a) adjacent-only action validity (per-agent home position h_i^{home} with a ring-distance- ≤ 1 constraint on a_i^{house}); (b) continuous-extinguish (each WORK at a BURNING house adds increment σ to a per-house accumulator; the fire transitions to SAFE deterministically when the accumulator reaches 1; fires do not burn out); (c) two-phase signalling (each night becomes two micro-rounds with round-1 emitting signal only and round-2 observing everyone’s round-1 signal); (d) distance-cost asymmetry (work cost augmented to $c + \alpha \cdot d(h_i^{\text{home}}, a_i^{\text{house}})$ where d is ring-arc distance, breaking ring-rotation symmetry). These variants are out of scope for the paper but are part of the broader calibrated family.

A.6 Relationship to canonical game-theory templates

Bucket Brigade recovers, as limiting cases or restrictions, several textbook templates.

Volunteer’s Dilemma. At $N=1, H=1$ with $\rho > 0, \beta=0$, each night the single agent faces a binary choice: WORK (pay c , save the house with probability κ) or REST (gain c_{rest} , lose the house in the next phase). This is the textbook single-shot Volunteer’s Dilemma. The N -player generalisation at $N > 1, H = 1$ is the *N -player Volunteer’s Dilemma*: the team needs at least one volunteer per fire, with the free-rider incentive equal to the $c + c_{\text{rest}}$ gap.

N -Player Public Goods Game. With $r_{\text{own}} = r_{\text{other}}$ and $p_{\text{own}} = p_{\text{other}}$ (ownership weights collapsed to uniform), the per-house ownership term reduces to a non-excludable common good and the team term is paid equally regardless of contribution. The marginal private return to WORK is then $\kappa \cdot (R_{\text{team}} + P_{\text{team}})/H - c$, which can be strictly less than the social marginal return $\kappa \cdot (R_{\text{team}} + P_{\text{team}}) - c$, producing the textbook public-goods underinvestment gradient.

Stag Hunt. In the limit $\kappa \rightarrow 1$ a single worker is sufficient to extinguish any fire, but the work cost c remains. The game then has two natural NE: the all-stag cooperative NE where every agent works coordinated houses (high payoff, requires coordination beliefs) and the all-hare lazy NE where everyone REST-defects (low payoff, no coordination required). The intermediate- κ regime interpolates between Stag Hunt and the public-goods regime.

Free-rider problem (rest-trap). At specific parameter cells the unique pure-strategy NE is asymmetric: $N-1$ agents play REST and one agent plays full-force WORK. This is the textbook free-rider problem in the bucket-brigade vocabulary; the phase diagram across (β, κ, c) shows the boundary between symmetric-NE-dominant and asymmetric-NE-dominant regimes.

Stochastic Game. Stripped to its bones, Bucket Brigade is a finite-state, finite-action, discounted stochastic game with N players, symmetric perfect monitoring, and reward decomposition into private and team components. Every formal result for finite stochastic games—minimax, equilibrium existence in stationary mixed strategies, value iteration convergence—applies directly. The contribution of the Bucket Brigade design is not in the game-theoretic class but in the *size*: at the minimal parameterisation the game is small enough that standard tools actually compute, which is the property the paper exploits.

A.7 Notation summary

Reproducibility note. The seven-phase ordering of §A.3 is load-bearing: implementers must apply phases in the exact order (signal write, location/mode write, extinguish, burn-out, spread,

Symbol	Meaning
$\mathcal{I} = \{1, \dots, N\}$	Agent set; N = population size
$\mathcal{H} = \mathbb{Z}/H\mathbb{Z}$	House ring; H = number of houses
S, B, R	House statuses: SAFE, BURNING, RUINED (R absorbing)
β	Per-night, per-neighbour fire-spread probability
κ	Solo-agent extinguish probability (independent-workers model)
ρ	Per-night, per-SAFE-house spontaneous-ignition probability
c	Per-night work cost (paid when $a_i^{\text{mode}} = 1$)
c_{rest}	Per-night rest payoff (implementation default 0.5)
$T_{\min}$	Minimum nights before termination allowed
$a_{t,i} = (a^{\text{house}}, a^{\text{mode}}, a^{\text{sig}})$	Per-agent action: target house, work bit, signal bit
$w_h(a)$	Number of WORKERS at house h under joint action a
$R_{\text{team}}, P_{\text{team}}$	Team reward / penalty per fraction-safe / fraction-ruined
$r_{\text{own}}, r_{\text{other}}$	Per-agent save-reward weights (own / other-owned house)
$p_{\text{own}}, p_{\text{other}}$	Per-agent ruin-penalty weights (own / other-owned house)
$n_S(t), n_R(t)$	Counts of SAFE / RUINED houses at end of night t
$\zeta_t \in \{0, 1\}^N$	Vector of broadcast signals (observable next night)
$\ell_t \in \mathcal{H}^N$	Vector of agent locations after phase 2

Table 3: Notation summary. Every downstream section (§3 equilibrium analysis, §4 experiments, §6 discussion, Appendix B NE derivation) reuses this vocabulary without redefinition.

spontaneous ignition, reward). The independent-workers extinguish formula of §A.3 and the post-extinguish, pre-burn-out spread test are the two details most often misread on first encounter; both are explicit above to make reimplementing tractable. A reader who reimplements from this appendix alone—using any RNG and any programming language—should obtain matching equilibrium structure and policy-evaluation results at the documented parameter cells.

B Analytical NE characterisation for the 4-agent game

This appendix gives the full derivation of the closed-form boundary inequalities reported in §3 and the predicted-vs-empirical comparison table referenced in Figures 1 and 2. The derivation reduces the $H=10$ ring with $T_{\min}=12$ to a one-house, four-agent, single-night stage game whose payoffs absorb multi-step contagion and ignition dynamics into a single survival coefficient. The three NE structures fall out as algebraic inequalities in $(\kappa, \bar{A}, c_{\text{gap}})$.

B.1 Setup

The analysis is restricted to the `minimal_specialization` parameter cell with $H=10, N=4, T_{\min}=12$ and reward tuple $W = (R_{\text{team}}, P_{\text{team}}, r_{\text{own}}, r_{\text{other}}, p_{\text{own}}, p_{\text{other}}) = (10, 10, 50, 0, 100, 0)$ at $\rho=0.02$. The free parameters are (β, κ, c) .

The analytical reduction is *not* a derivation of the full stochastic-game subgame-perfect equilibrium. The full game’s NE conditions exist only via numerical value iteration on the $\geq 9 \times 10^9$ -state space, which does not generalise across (β, κ, c) in closed form. Restricting to the per-night stage game under stationary opponent strategies is the load-bearing simplification that makes algebraic boundary inequalities possible; §B.5 reports where this restriction produces visibly wrong predictions.

Mapping to the empirical strategy space. The empirical phase diagram is computed by the heterogeneous Double-Oracle solver over continuous 10-D heuristic-parameter vectors, not the binary WORK/REST stage game the derivation uses. The bridge is the work-tendency parame-

ter: Hero (`work_tendency=1.0`) maps to analytical all-WORK; Firefighter (`work_tendency=0.9`) is treated as near all-WORK; FreeRider (`work_tendency=0.2`, `rest_reward_bias=0.9`) maps to all-REST. Empirical verdicts map to analytical NE structures as: `symmetric_only` with profile $4 \times \text{Hero} \rightarrow \text{symmetric all-WORK}$; `asymmetric_only` with profile $1 \times \text{Hero} + 3 \times \text{Firefighter} \rightarrow \text{asymmetric 1-WORKER}$; `no_convergence` $\rightarrow$ no-pure-NE collapse. The mapping is approximate—Firefighters at `work_tendency=0.9` are not literal REST agents—and the quantitative bias this introduces is named in §B.5.

Single-house mean-field reduction. Instead of analysing the full 10-house ring with 4 agents distributed across houses, we collapse the analysis to a representative single house with all 4 agents present, where the night’s outcome is paid out as the expected per-agent payoff under the stage-game profile. Per-night reward decomposition follows §A.4: a work cost $-c$ on WORK and rest payoff $+c_{\text{rest}}=0.5$ on REST; a per-house ownership term weighted by $r_{\text{own}}=50$ on save and $p_{\text{own}}=100$ on every RUINED night; and a team term scaled by $R_{\text{team}}=P_{\text{team}}=10$. In the single-house reduction ownership accrues to the house’s designated owner and is shared equally across agents in the steady-state assumption that each agent owns one house and the rotation is symmetric.

B.2 Mean-field stage-game reduction

Survival coefficient. With $k \in \{0, 1, 2, 3, 4\}$ workers at the house on a given night, the probability of extinguish in phase 3 is $1 - (1 - \kappa)^k$. Under the canonical phase order the spread phase has no BURNING source houses to spread from, so β does not enter the leading-order single-night payoff and contagion enters only via spontaneous ignition at rate ρ per SAFE house. Let s be the probability a house is SAFE; conditional on SAFE at night t and k workers on night $t+1$, the per-night ruin probability is

$$q(k) := 1 - \Pr[s_{t+1}=S \mid s_t=S, k] = \rho \cdot (1 - \kappa)^k.$$

Over $T_{\text{min}}=12$ nights the probability the house survives is approximately $(1 - q(k))^{T_{\text{min}}} \approx 1 - T_{\text{min}} \cdot q(k)$ for small $q(k)$. Collecting the $q(k)$ -dependent terms of the expected per-house reward yields

$$\mathbb{E}[\text{per-house reward loss} \mid k] = A \cdot \rho \cdot (1 - \kappa)^k,$$

with

$$A = r_{\text{own}} \cdot T_{\text{min}} + p_{\text{own}} \cdot T_{\text{min}} + P_{\text{team}} \cdot T_{\text{min}}/H = 50 \cdot 12 + 100 \cdot 12 + 10 \cdot 12/10 = 1812.$$

The marginal benefit of adding the k -th worker (going from k to $k+1$) is the reduction in expected loss,

$$\Delta(k \rightarrow k+1) = A \cdot \rho \cdot [(1 - \kappa)^k - (1 - \kappa)^{k+1}] = A \cdot \rho \cdot \kappa \cdot (1 - \kappa)^k.$$

Writing $\tilde{A} := A\rho = 1812 \cdot 0.02 = 36.24$ for the per-night survival coefficient, the marginal benefit at the k -th worker is $\tilde{A} \cdot \kappa \cdot (1 - \kappa)^k$.

Stage-game payoffs. Each agent picks $a^{\text{mode}} \in \{0, 1\}$. With k_{-i} other workers under the joint profile, agent i ’s per-night payoff difference between WORK and REST is

$$u_i(\text{Work} \mid k_{-i}) - u_i(\text{Rest}) = \tilde{A} \cdot \kappa \cdot (1 - \kappa)^{k_{-i}} - c_{\text{gap}},$$

with $c_{\text{gap}} := c + c_{\text{rest}}$. For the phase-diagram driver’s $c=0.5$ and the default $c_{\text{rest}}=0.5$, $c_{\text{gap}} = 1.0$.

Ring locality (statement). The single-house reduction assumes all 4 agents are at the same house. The empirical setup spreads 4 agents across 10 houses each night; under a symmetric uniform policy, each house sees $E[k] = N/H = 0.4$ workers per night. The derivation in §B.3 keeps the single-house framing (all 4 agents at one fire) because that is the case that determines whether a “1 Worker, 3 free-riders” profile is stable; the ring-locality correction is the perturbation in §B.5.

B.3 The three NE candidates

Symmetric all-Work NE. Under the all-WORK profile, every agent plays $a_i = \text{Work}$. For this to be a NE, a unilateral deviation to REST must not strictly increase payoff. The deviation comparison has $k_{-i}=3$:

$$\tilde{A} \cdot \kappa \cdot (1 - \kappa)^3 \geq c_{\text{gap}} \quad (\text{S}).$$

Substituting $\tilde{A}=36.24$, $c_{\text{gap}}=1.0$: $f(\kappa) := \kappa(1 - \kappa)^3$ is unimodal on $[0, 1]$ with maximum $f(1/4) = 27/256 \approx 0.105$. Solving $f(\kappa) = 0.0276$ numerically gives roots $\kappa \approx 0.030$ and $\kappa \approx 0.65$; the symmetric NE exists for $\kappa \in [0.030, 0.65]$.

Asymmetric 1-Worker NE. Under the asymmetric profile exactly 1 agent WORKS and 3 REST. Two conditions: the lone WORKER prefers WORK ($k_{-i}=0$) and each free-rider prefers REST ($k_{-i}=1$):

$$\tilde{A} \cdot \kappa \geq c_{\text{gap}} \quad \text{AND} \quad \tilde{A} \cdot \kappa \cdot (1 - \kappa) < c_{\text{gap}} \quad (\text{A}).$$

Condition (a): $36.24 \cdot \kappa \geq 1.0 \Leftrightarrow \kappa \geq 0.028$. Condition (b): $g(\kappa) := 36.24\kappa(1 - \kappa)$ is unimodal with $g(1/2) = 9.06$; the roots of $g(\kappa) = 1.0$ are $\kappa \approx 0.028$ (lower) and $\kappa \approx 0.972$ (upper). Condition (b) requires κ outside $[0.028, 0.972]$. The intersection of (a) and (b) is essentially empty below $\kappa=0.028$ and gives the high- κ band $\kappa \gtrsim 0.972$, the Stag-Hunt-like regime where a single worker is overwhelmingly likely to extinguish and the free-rider’s marginal contribution is negligible.

No-pure-NE collapse. The all-REST profile is never a pure-strategy NE for any $\tilde{A} > 0$ such that $\tilde{A}\kappa > c_{\text{gap}}$, because a single deviation to WORK earns $\tilde{A}\kappa - c_{\text{gap}} > 0$. The no-pure-NE collapse regime is the parameter regime where neither (S) nor (A) holds AND the all-WORK profile is dominated. From the above, $\tilde{A}\kappa < c_{\text{gap}}$ implies the lone-WORKER deviation does not pay (A (a) fails) and a fortiori (S) fails (since $(1 - \kappa)^3 < 1$ for $\kappa > 0$):

$$\tilde{A} \cdot \kappa < c_{\text{gap}} \implies \text{no pure-strategy NE (collapse)} \quad (\text{C}).$$

Summary table. For the `minimal_specialization` parameter cell at $c=0.5$ (so $c_{\text{gap}}=1.0$), $\rho=0.02$, $T_{\text{min}}=12$, and $\tilde{A} = 36.24$:

κ range	Symmetric (S)	Asymmetric (A)	Collapse (C)	Predicted verdict
$\kappa < 0.028$	No	No	Yes	collapse (no-pure-NE)
$0.028 \leq \kappa \leq 0.030$	Borderline	Empty	No	borderline; may be mixed
$0.030 \leq \kappa \leq 0.65$	Yes	No	No	symmetric_only
$0.65 < \kappa \leq 0.972$	No	No	No	mixed or transition
$\kappa > 0.972$	No	Yes	No	asymmetric_only

Table 4: Predicted NE structure under the mean-field reduction at $c=0.5$, $\rho=0.02$, $T_{\text{min}}=12$. The qualitative phase order from κ -small to κ -large is **collapse** $\rightarrow$ **symmetric_only** $\rightarrow$ **mixed/no-pure** $\rightarrow$ **asymmetric_only**, matching the empirical ordering on the 39-cell grid at $c=0.5$. β does not enter under the canonical phase order.

B.4 Predicted vs. empirical phase table (7-cell preview)

The 7-cell preview at $c=0.5, \rho=0.02$ against the predicted verdict:

β	κ	Empirical verdict	Predicted verdict	Match?
0.10	0.10	no_convergence	symmetric_only (in 0.030–0.65 band)	×
0.50	0.10	no_convergence	symmetric_only	×
0.90	0.10	no_convergence	symmetric_only	×
0.50	0.50	symmetric_only	symmetric_only	✓
0.90	0.50	symmetric_only	symmetric_only	✓
0.50	0.90	asymmetric_only	mixed/borderline (predicted threshold 0.972)	partial
0.90	0.90	asymmetric_only	mixed/borderline	partial

Table 5: Predicted vs. empirical NE verdict on the 7-cell preview reported in §3 of the body. Qualitative-vs-quantitative split: β -independence holds 3/3 on κ -rows with both β values; the qualitative phase ladder `no_convergence` $\rightarrow$ `symmetric_only` $\rightarrow$ `asymmetric_only` matches; exact cell-by-cell agreement is 2/7 because the predicted κ thresholds are wrong. The predicted collapse threshold $\kappa=0.028$ is 3–4 $\times$ too low; the predicted asymmetric-NE threshold $\kappa=0.972$ is $\sim 10\%$ too high. The full 39-cell grid in §3 of the body widens the sample without changing the qualitative findings.

B.5 Where the reduction breaks

Five sources of systematic bias are identifiable from the derivation.

Ring locality (the $H=10 \rightarrow$ mean-field gap). The reduction puts all 4 agents at one house. Empirically, 4 agents distribute across 10 houses each night; under a symmetric uniform policy, each house sees $E[k] = 0.4$ workers per night on average rather than 4. The effective per-house extinguish probability is $1 - (1 - \kappa)^{0.4}$ rather than $1 - (1 - \kappa)^4$ —a much weaker defence. For $\kappa=0.5$: full-mean-field gives $1 - 0.5^4 = 0.9375$; the ring-corrected average gives $1 - 0.5^{0.4} \approx 0.24$. The ring-corrected $\tilde{A}$ is therefore much smaller than 36.24, and the collapse boundary κ at which $\tilde{A}\kappa < c_{\text{gap}}$ shifts substantially upward. Predicted correction: the collapse threshold moves from $\kappa=0.028$ (single-house) to roughly $\kappa \in [0.1, 0.3]$ (ring-corrected), consistent with the empirical boundary. This is the dominant systematic bias; tightening it without losing tractability is the open problem.

Heuristic strategy space (Firefighter $\neq$ all-Rest). The empirical solver picks **Firefighters** (`work_tendency=0.9`) as the “free-riders” in the asymmetric NE profile, not literal **RESTers** (`work_tendency=0.2`). Three **Firefighters** at 90% WORK contribute non-zero expected extinguish probability—the effective k_{-i} in the lone-WORKER’s deviation comparison is closer to $1 + 3 \cdot 0.9 = 3.7$ than to the analytical $k_{-i}=1$. At $\kappa=0.9$, $\tilde{A}\kappa(1 - \kappa)^{3.7} \approx 0.013 \ll c_{\text{gap}}=1.0$, so the lone WORKER should not exist as a NE under that interpretation. The empirical solver still reporting `asymmetric_only` at $\kappa=0.9$ suggests the per-agent ownership reward—not captured in the single-house mean-field reward—anchors each agent to its own house even when free-riding on others’ fires. The asymmetric NE is therefore partly an artefact of per-agent ownership, not a pure Volunteer’s-Dilemma equilibrium.

Per-agent ownership rewards (the W gap). The analytical A coefficient lumps $r_{\text{own}}=50$ and $p_{\text{own}}=100$ into the survival term, but the per-agent reward only fires on the agent’s *own* house. The single-house reduction assumes the representative house is the agent’s own house with probability

1; the ring-corrected reality is probability $1/H=0.1$. The other 9 houses contribute through the team term only, which scales A down by a factor of roughly $(p_{\text{own}}/H + P_{\text{team}}/H)/p_{\text{own}} \approx 0.11$ on the off-own-house terms. The ring-corrected A sits between $A/H=181$ and $A=1812$, depending on the agent’s house-target distribution.

β -independence as leading-order, not exact. The reduction’s β -independence is exact under the canonical phase order (phase 5 has no BURNING sources). The phase-diagram driver uses canonical mode, so this matches. But empirically β may enter via the across-night dynamics: a high- β regime makes neighbour-driven ignition the dominant failure mode over multiple nights, even with within-night spread inert. The reduction’s leading-order claim “ β does not enter the single-night payoff comparison” is exact; the second-order claim “ β does not enter multi-night survival” is approximately true under canonical phase order and small $\beta\rho$ products, but should break for $\beta\rho \gg \rho$. The current grid at $\rho=0.02$ is the wrong regime to test this; sweeping $\rho \in \{0.05, 0.10\}$ would let the framework be falsified.

Single-stage vs. multi-stage SPE. The reduction is the single-night stage game under stationary opponent policies, not the full multi-stage subgame-perfect equilibrium. Empirically, the DO solver evaluates policies over full $T_{\min}=12$ episodes, so per-cell payoffs reflect the integrated multi-stage outcomes. A symmetric all-WORK profile that is a NE of the single-night stage game may fail to be a NE of the multi-stage game if a deviator can sustain a long-horizon advantage by switching to REST at a specific night. The reduction folds multi-stage dynamics into A by integrating expected survival over $T_{\min}$ nights, but the exact multi-stage NE conditions involve subgame-perfect best-response calculations the single-night reduction cannot represent.

B.6 Implications and recommended follow-ups

Even with the quantitative thresholds wrong by 3–10 $\times$, the three structural predictions the mean-field reduction makes—(a) κ -monotonicity of the verdict, (b) β -independence under canonical phase order, (c) the existence of a no-pure-NE collapse regime at small κ —are all empirically observed and would not be derivable without the closed-form derivation. The body of the paper adopts inequalities (S), (A), (C) as the analytical anchor.

Three concrete follow-ups, in priority order: (1) complete the empirical grid across $c \in \{0.5, 1.0, 2.0\}$ and re-run the comparison table (now done at the 39-cell scale of §3); (2) sweep $\rho \in \{0.05, 0.10\}$ at fixed $c=0.5$ to test the β -independence breakage predicted in §B.5; (3) derive a ring-Markov correction to A that replaces the single-house mean-field assumption with a chain-Markov approximation over the 10-house ring, closing the §B.5 gap without paying the full state-space cost.